

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

1st Semester of M. Pharm (Clinical Pharmacy)

University Theory Examination December 2015

PH813.03 Advance Pharmacokinetics & Clinical Applications

Date: 21/12/2015 , Monday

Time: 10.00 a.m to 1.00 p.m

Maximum Marks: 70

Instructions:

1. Q.1 and Q.4 are compulsory.
2. Figures to right indicate marks.
3. Section wise separate answer book to be used.
4. Draw neat sketches wherever necessary.

SECTION-I

- Q.1** **A)** Describe objectives & functions of Therapeutic Drug monitoring (TDM) system **[05]**
- B)** Describe Pharmacokinetic Correlation in drug therapy **[05]**
- Q.2** **A)** Describe significant factors for Conversion from intravenous to oral dosing **[05]**
- B)** Explain: How to determine the dose of the drugs and dosing intervals? **[05]**

OR

- Q.2** **A)** Describe Indications and Protocols for TDM. **[05]**
- B)** Explain: How to calculate the dose in the elderly and pediatrics populations? Why? **[05]**
- Q.3** **Answer the following: (Any three)** **[15]**
- A)** Describe the significance of TDM of drugs in relation to specific disease / condition
- B)** Describe clinical significance of Enzyme Inhibition and Induction
- C)** Describe Pharmacokinetic drug interactions with examples
- D)** Which are the variables affecting individualized drug dosage regimen? Explain in brief in terms of TDMD OR Describe the significance of plasma half life with examples.

P.T.O

SECTION-II

Q.4 Explain in detail objectives of bio-availability & bioequivalence studies. Explain measurements methods of bio-availability. [10]

Q.5 Explain followings in brief:

- (A) Explain dose adjustment methods in Renal impairment condition [4]
- (B) Measurement of Glomerular Filtration rate and creatinine clearance rate [3]
- (C) Effect of Hepatic disease on pharmacokinetics of dosage adjustment [3]

OR

Q.5 (A) Explain General approach for dosage adjustment in Renal disease [4]

(B) Explain stem cell – a new therapeutic approach [4]

(C) Provide brief note on Pharmacogenetics [2]

Q.6 **Explain followings (Any three)** [15]

(A) Provide brief note on experimental Designs in Bioequivalence study

(B) Drug dosing in pregnancy and lactation

(C) Explain clinical view points and future perspectives of Pharmacogenetics

(D) A short note on regulatory aspects of bio availability and bioequivalence studies